

City General Hospital

30-Day Readmission Risk Assessment

Patient ID:	20251103-COP-81174c	Generated:	November 03, 2025 11:48
Patient Name:	N/A	Age:	36 years
Sex:	M	Disease:	COPD

30-Day Readmission Risk: **34.1%** | Classification: **LOW RISK** (Threshold: 50%)

Index Admission Summary

Length of Stay:	9 days	Discharge Destination:	home_with_care
Prior Admissions (90d):	0	Comorbidities Count:	0
Follow-up Scheduled:	Yes	Admission Type:	N/A

Top Contributing Risk Factors (SHAP Analysis)

Feature	Value	Contribution	Impact	Interpretation
prior_admissions_90d	0.00	-1.222	↓ Lower Risk	prior_admissions_90d influences readmission risk.. This feature has a major effect and reduces the readmission risk.
Use_of_home_oxygen	0.00	-0.766	↓ Lower Risk	Use_of_home_oxygen influences readmission risk.. This feature has a moderate effect and reduces the readmission risk.
Oxygen_saturation	92.03	+0.525	↑ Higher Risk	Low oxygen indicates inadequate ventilation or exacerbation. (92.03%; ref 92-100%). This feature has a moderate effect and increases the readmission risk.
FEV1_percent	64.77	-0.287	↓ Lower Risk	FEV1_percent influences readmission risk.. This feature has a minor effect and reduces the readmission risk.
length_of_stay	9.00	+0.248	↑ Higher Risk	length_of_stay influences readmission risk.. This feature has a minor effect and increases the readmission risk.
followup_scheduled	1.00	-0.245	↓ Lower Risk	followup_scheduled influences readmission risk.. This feature has a minor effect and reduces the readmission risk.
Prior_exacerbations_1yr	1.00	+0.223	↑ Higher Risk	Prior_exacerbations_1yr influences readmission risk.. This feature has a minor effect and increases the readmission risk.
sex	M	+0.134	↑ Higher Risk	sex influences readmission risk.. This feature has a minimal effect and increases the readmission risk.

Clinical Summary

This 36 years m patient presents with a **low risk 30-day readmission risk (34.1%)** following discharge for COPD management. The risk assessment threshold for this condition is 50%, placing this patient below the high-risk threshold.

Primary Risk Drivers: prior_admissions_90d, Use_of_home_oxygen, Oxygen_saturation are identified as major contributing factors.

Recommendation: Close follow-up and aggressive management of identified risk factors is recommended to prevent readmission.

Clinical Management Recommendations

- **Primary Disease Management:** Review and optimize current treatment plan for COPD. Consider consultation with appropriate specialists.
- **Medication Reconciliation:** Complete medication review at discharge. Ensure patient understands all medications, dosing, and timing.
- **Care Coordination:** Schedule follow-up appointment within 7 days of discharge. Consider home health services for high-risk patients.
- **Patient Education:** Reinforce medication adherence, warning signs requiring immediate attention, and lifestyle modifications.
- **Laboratory Monitoring:** Schedule appropriate lab work based on disease-specific guidelines and medication monitoring requirements.

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Medication Recommendations

Patient ID:	20251103-COP-81174c	Patient Name:	N/A
Age / Sex:	36 years / M	Disease:	COPD

Important: The following medication recommendations are based on current clinical guidelines and the patient's risk profile. All medications must be reviewed, prescribed, and adjusted by the attending physician based on individual patient factors, allergies, drug interactions, and institutional protocols.

Recommended Medication Protocol:

Bronchodilators:

- LABA + LAMA combo (e.g., Tiotropium + Olodaterol inhaler)
- Short-acting beta-agonist (Albuterol) as rescue inhaler
- Theophylline if persistent symptoms

Anti Inflammatory:

- Inhaled corticosteroids (e.g., Fluticasone) if frequent exacerbations
- PDE4 inhibitor (Roflumilast) for severe COPD
- Oral corticosteroids (Prednisone 40mg x 5 days) during exacerbations

Oxygen Therapy:

- Long-term oxygen if SpO2 <88% or PaO2 <55 mmHg
- Pulmonary rehabilitation program
- Smoking cessation support (Varenicline or Bupropion)

Monitoring:

- Spirometry annually, pulse oximetry, vaccination (flu + pneumococcal)

Note: These are general guidelines. All medications should be prescribed and adjusted by the attending physician based on individual patient factors.

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Potential Disease Progression & Related Conditions

Patient ID:	20251103-COP-81174c	Patient Name:	N/A
Age / Sex:	36 years / M	Disease:	COPD

Potential Disease Progression & Related Conditions:

[HIGH RISK] High-Risk Conditions (Requires Active Prevention):

1. Respiratory Failure

- *Risk Factors:* FEV1 <30%, frequent exacerbations, hypoxemia
- *Typical Time Frame:* 2-5 years
- *Prevention Strategy:* Oxygen therapy, pulmonary rehab, medication adherence

2. Cor Pulmonale (Right Heart Failure)

- *Risk Factors:* severe COPD, chronic hypoxia, pulmonary hypertension
- *Typical Time Frame:* 5-10 years
- *Prevention Strategy:* Long-term oxygen, diuretics, treat underlying COPD

3. Lung Cancer

- *Risk Factors:* smoking history, age >55, COPD severity
- *Typical Time Frame:* 10-20 years
- *Prevention Strategy:* Smoking cessation, annual low-dose CT screening

[MODERATE RISK] Moderate-Risk Conditions (Monitor Closely):

1. Pneumonia (Recurrent)

- *Risk Factors:* impaired clearance, immunosuppression
- *Prevention:* Vaccination, inhaler technique, smoking cessation

Personalized Risk Assessment:

- Current markers within acceptable ranges. Continue monitoring.

Disclaimer: This report is generated by a machine learning model for clinical decision support. All recommendations should be reviewed and approved by qualified healthcare professionals. Medication dosages and treatment plans must be individualized based on patient-specific factors, comorbidities, and current clinical guidelines.